

Spectral Flow in the Unified Phase of the Parallel Key Geometric Flow

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Abstract

This paper examines the behavior of spectral flow associated with the unified operator evolution in the Parallel Key Geometric Flow (PKGF). The PKGF framework combines a conservative commutator-type component with a dissipative component generated by an elliptic operator, integrated through complexification. When the resulting operator family is realized as a perturbation of a fixed elliptic operator, it naturally falls within the scope of classical spectral flow theory for self-adjoint Fredholm operators.

The purpose of this work is not to introduce new results in spectral flow theory, but to clarify (i) how PKGF-generated operator families relate to the classical definition of spectral flow, (ii) the analytical conditions—Fredholmness, norm-resolvent continuity, and transversality—under which spectral flow is well-defined, and (iii) the qualitative roles of the conservative and dissipative components in eigenvalue motion. The results are intended as a technical supplement to the PKGF framework, providing a bridge between its unified dynamics and established results in elliptic operator theory.

Keywords: Parallel Key Geometric Flow, PKGF, Spectral Flow, Unified Phase, Fredholm Operators, Operator Evolution

1 Introduction

The Parallel Key Geometric Flow (PKGF) defines a linear evolution equation that combines a conservative component generated by commutator action $[\Omega, K]$ and a dissipative component generated by an elliptic operator $\mathcal{D}$, integrated via complexification $\tilde{K} = K_{\text{core}} + iK_{\text{fluct}}$.

This unified evolution produces a time-dependent family of operators $\tilde{K}(t)$. When these operators are used as zeroth-order perturbations of a fixed self-adjoint elliptic operator $\mathcal{L}_0$, one obtains the family

$$\mathcal{L}(t) = \mathcal{L}_0 + \tilde{K}(t).$$

The present paper aims to clarify how this family interacts with classical spectral flow theory. No new results in spectral flow theory are claimed; the focus is on the proper application of existing theory to the PKGF setting.

2 Mathematical Setting and Assumptions

We adopt the structural axioms (A1–A6) and analytical hypotheses (HmD, Hdom, Hreg, HFred, SFop) from the foundational PKGF paper.

Let M be a compact Riemannian manifold, $E \rightarrow M$ a finite-rank vector bundle, $\tilde{K}(t) \in \Gamma(\text{End}(E))$ the unified operator, and $\mathcal{L}(t) = \mathcal{L}_0 + \tilde{K}(t)$ its elliptic realization, where $\mathcal{L}_0$ is self-adjoint and elliptic.

We assume:

- **H-Fred:** $\mathcal{L}(t)$ remains Fredholm for all t ,
- **SF-Op:** the family $\{\mathcal{L}(t)\}$ is norm-resolvent continuous,
- **Transversality:** $\dot{\lambda}_j(t_c) \neq 0$ at simple crossing times t_c .

These assumptions are standard in the spectral flow literature (Phillips 1996; Waterstraat 2016; Bär–Ziemke 2025; Booss-Bavnbek et al. 2005; Doll et al. 2023).

3 Spectral Flow for PKGF Operator Families

Under the above assumptions, the spectral flow

$$\text{SF}(\mathcal{L}(t)) \in \mathbb{Z}$$

is well-defined in the classical sense (Atiyah et al. 1969; Phillips 1996). The PKGF evolution does not modify the definition or properties of spectral flow; it provides a specific class of time-dependent perturbations to which the theory naturally applies. The stability of the spectral flow under complexified perturbations is ensured by the gap topology of unbounded Fredholm operators (Booss-Bavnbek et al. 2005; Doll et al. 2023). Furthermore, the topological change in the unified phase can be related to the emergence of dimensional structures via the Atiyah–Patodi–Singer index theorem (Melrose 1993; Van den Dungen & Ronge 2020).

4 Eigenvalue Dynamics

For a simple eigenvalue $\lambda_j(t)$ of $\mathcal{L}(t)$ with normalized eigenvector $u_j(t)$, the Hellmann–Feynman-type formula gives

$$\dot{\lambda}_j(t) = \langle \dot{\mathcal{L}}(t) u_j(t), u_j(t) \rangle,$$

provided sufficient regularity holds (Waterstraat 2016).

Since

$$\dot{\mathcal{L}}(t) = [\Omega, \tilde{K}(t)] + \lambda \mathcal{D}(\tilde{K}(t)),$$

both the conservative and dissipative terms may contribute to eigenvalue motion. In many settings, the dissipative term contributes to the real part of the variation, while the conservative term may introduce rotational effects. A precise quantitative decomposition in infinite dimensions requires careful perturbation-theoretic analysis (Davies 2006).

5 Main Proposition

5.1 Proposition (Spectral flow for PKGF-generated operator families)

Under assumptions HFred, SFOp, and the transversality condition $\dot{\lambda}_j(t_c) \neq 0$ at each simple crossing time t_c , the spectral flow is given by

$$\text{SF}(\mathcal{L}(t)) = \sum_{t_c} \text{sign}(\dot{\lambda}_j(t_c)).$$

This is a direct application of standard spectral flow theory (Phillips 1996; Waterstraat 2016).

5.2 Remark.

This proposition is included to clarify how PKGF-generated operator families fit into the classical framework. It does not assert novelty in spectral flow theory.

6 Discussion

The unified phase of PKGF generates a natural class of time-dependent perturbations of elliptic operators. While it does not introduce new spectral flow phenomena, it provides a structured context in which conservative (symmetry-preserving) and dissipative (energy-damping) dynamics coexist and may induce eigenvalue crossings when transversality holds.

The interplay between the two components in infinite dimensions is non-trivial and merits further study using more refined perturbation theory. Extensions to higher-order spectral flow (Dai & Zhang 1998) or lattice-based K-theoretic computations (Aoki et al. 2025) may provide insights into the discretization of these flows.

7 Conclusion

This paper provides a technical clarification of how classical spectral flow theory applies to the unified phase of the Parallel Key Geometric Flow. By identifying the relevant analytical conditions and relating PKGF dynamics to established results in operator theory, it serves as a bridge between the PKGF framework and classical elliptic operator theory.

8 Numerical Evidence: Spectral Flow and Correlation in Chaotic Dynamics

Numerical experiments using the Logistic Map and the Lorenz system via Noetics SDK v1.0 have confirmed the physical reality of the theorized Spectral Flow and "Rank Jumps."

8.1 Phase Coherence and Structural Predictive Capacity

In the Lorenz system under dynamic gauge control, the geometric integrity of the attractor was maintained long after coordinate-level predictions (RMSE) failed. Within the operator family $\mathcal{L}(t) = \mathcal{L}_0 + \tilde{K}(t)$, the period where major eigenvalue pairs avoid zero-crossing (prevention of catastrophic Spectral Flow) corresponds to the manifestation of **Structural Predictive Capacity**. The persistent phase coherence of eigenvalues suggests that the spectral flow remains within a topologically "invariant" regime during this period, enabling the system to prophesy macroscopic structural truths even when microscopic trajectories become unpredictable.

8.2 Direct Observation of Rank Jumps

Discontinuous changes in effective dimension d_{eff} (Rank Jumps) were observed during noise removal in the Logistic Map and at the point of structural collapse in the Lorenz system ($t = 1.425\text{s}$). This event coincided with a massive Spectral Flow where major eigenvalues dropped below the threshold, corroborating the view that phase transitions in intelligence are "topological discontinuities" in eigenvalue dynamics rather than a mere "cumulative increase in continuous error."

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